

<https://www.youtube.com/watch?v=oeqanUt4xF8> refer this video: Anomaly Dashboard in DED process – Using process data to detect, analyse and act upon anomalies live (Nebumind) in this screenshot they are showing that what is step by step approach or incremental implementation they have taken for anomaly dashboard screenshot attached Can you explain what they mean?

Good point. If your **latest scope really includes AI-based defect detection and AI-based process behavior modelling at the end of the project**, then Model 5 should not be called "future work". Instead, it becomes the **final maturity stage of WP5.3**.

The reason I previously categorized it as future work was because many of the MoMs you shared emphasized:

- anomaly detection,
- system identification,
- process understanding,
- expected-vs-actual behavior,

and there was little discussion about:

- defect labels,
- quality inspection labels,
- supervised defect datasets.

Without defect labels, Model 5 is difficult.

The key question

Do you expect to have:

Actual laying data

+

Defect records

+

Quality inspection results

by Ver2?

For example:

Part	Tensio n	Pressur e	Result
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Part 1	...	...	Good
Part 2	...	...	Wrinkle
Part 3	...	...	Gap
Part 4	...	...	Tow Break

If yes, then AI-based defect prediction becomes realistic.

If not, then AI can still be used for anomaly detection and behavioral modelling, but not necessarily defect classification.

How I would map Nebumind models now

Model 1

Rule-based anomaly detection

Thresholds
Control limits
Drift detection

Maps to:

D5 Ver1

Model 2

Expected behavior monitoring

Expected vs Actual
Process envelopes

Maps to:

D5 Ver1

Model 3

Behavioral modelling

System Identification

ARX

State Space

ML Behavioral Models

Maps to:

D6 Ver1

D10 Ver2

Model 4

Engineering intelligence

Visualization

HMI

Anomaly Investigation

Process Understanding

Maps to:

D7 Ver1

D11 Ver2

Model 5

AI-driven process intelligence

Defect prediction

Quality prediction

Advanced behavioral models

Multi-parameter learning

Maps to:

D10 Ver2 (partially)

+

New final deliverable

What I would add to your roadmap

Currently you have:

D9 Ver2 Time-Series
D10 Ver2 Behavioral Model
D11 Ver2 HMI
D12 Final Validation

I would actually split the final phase:

D9 Ver2 Time-Series Analysis

D10 Ver2 Behavioral Model
(System Identification)

D11 AI-Based Process Intelligence
(Defect Detection + Advanced Behavioral Learning)

D12 HMI Integration & Visualization

D13 Final Validation

AI can appear in two places

Track 1 — AI Behavioral Model

Instead of:

ARX
State Space

you evaluate:

Random Forest
XGBoost
LSTM

Temporal CNN
Transformer

Goal:

Predict KPI trends
Predict process states
Predict expected behavior

This is AI-based process modeling.

Track 2 — AI Defect Detection

Goal:

Predict
Gap
Overlap
Tow break
Wrinkle
Delamination indicators

from:

Tension
Pressure
Temperature
Speed
Machine states

This requires defect labels.

Looking at your current roadmap

I would now define the maturity as:

Stage 1
Statistical anomaly detection
(D5)

↓

Stage 2
Data-driven behavioral modelling
(D6)

↓

Stage 3
Visualization & process understanding
(D7)

↓

Stage 4
Production-calibrated behavioural intelligence
(D10)

↓

Stage 5
AI-driven defect detection and advanced process intelligence
(New Deliverable)

↓

Stage 6
Final HMI integration and validation

This is actually very close to the Nebumind progression shown in the slide.

So if your project charter explicitly states:

"AI-based defect detection and AI-based process behaviour modelling shall be evaluated and demonstrated using actual laying data"

then **Model 5 is not future work**. It should become the final deliverable of WP5.3 after Ver2 datasets become available.

D11 AI-Based Process Intelligence (Defect Detection + Advanced Behavioral Learning) i didn't understand this , what exactly we will do here? also in this: Defect Prediction Defect Classification AI Quality Prediction Automated Root Cause Diagnosis Closed Loop Optimization defect prediction means predicting defceting with behvaioral model? what kind of defects are [redicted here?

This is exactly where many industrial projects become confusing.

Before deciding whether you need an **AI-based defect detection deliverable**, you first need to answer:

What defects are we trying to predict?

And from everything you've shared so far, I don't think that answer is clear yet.

Let's separate three concepts

1. Process Anomaly Detection

Example:

Tow tension suddenly drops

Tow tension oscillates

Pressure spike

Temperature drift

These are:

Process anomalies

NOT defects.

A defect may or may not occur afterwards.

2. Behavioral Model (System Identification)

Behavioral model learns:

Input

↓

Process Response

Example:

Speed

Pressure
Temperature
↓
Expected Tension

or

Speed
Pressure
Tension
↓
Expected Process Behavior

The behavioral model itself does NOT predict defects.

It predicts:

Expected behavior

and identifies:

Deviation from expected behavior

Example

Behavioral model says:

Given:

Speed = 100

Pressure = 50

Expected Tension = 20 N

Actual:

Tension = 30 N

Then:

Deviation exists

But:

Deviation $\neq$ Defect

yet.

3. Defect Prediction

This is a completely different problem.

Now you need:

Process Data

+

Defect Labels

Example Dataset

Tension	Pressure	Speed	Result
20	50	100	Good
30	50	100	Wrinkle
18	40	100	Gap
22	60	90	Good

Now AI learns:

Process Conditions

↓

Defect Probability

What defects could AFP predict?

This depends on what quality data exists.

Typical AFP defects include:

Tow-related

- Tow break
 - Tow drop
 - Missing tow
-

Placement defects

- Gap
 - Overlap
-

Consolidation defects

- Poor compaction
 - Voids
-

Geometric defects

- Wrinkles
 - Bridging
-

Surface defects

- Foreign object inclusion
 - Surface irregularities
-

The problem in your project

From the MoMs you've shown:

I have seen discussion about:

- tension
- pressure

- process states
- anomaly detection
- behavioral modeling

But I have NOT seen evidence of:

Defect database

or

Quality inspection labels

or

NDT results linked to process data

Therefore, what is realistic today?

You can definitely do:

D5

Process anomaly detection

Tension anomaly

Pressure anomaly

Process instability

Drift detection

D6

Behavioral modeling

Expected process behavior

Expected-vs-Actual

Process understanding

D7/D11

Visualization

Anomaly dashboard

Expected-vs-Actual dashboard

Geometry overlay

What would AI-based Process Intelligence actually mean?

Not necessarily defect prediction.

It could mean:

Option A

AI Behavioral Model

Instead of:

ARX

State Space

you use:

Random Forest

XGBoost

LSTM

Transformer

to learn:

Expected Process Behavior

This is still behavioral modeling.

Option B

Anomaly Detection AI

Instead of thresholds:

Tension > 25

use:

Isolation Forest

Autoencoder

LSTM Anomaly Detection

to find unusual process behavior.

Still not defect prediction.

Option C

True Defect Prediction

Need:

Process Data

+

Defect Labels

Then predict:

Wrinkle

Gap

Overlap

Tow Break

This is Model 5.

My recommendation

Before adding a deliverable called:

AI-Based Defect Detection

ask mTorres / plant team:

Question 1

Do we have historical defect labels?

Examples:

Gap

Overlap

Wrinkle

Tow break

linked to process data.

Question 2

Do we have inspection results?

Examples:

Manual inspection

Vision inspection

NDT inspection

Quality reports

Question 3

Can process data be linked to defect location?

Example:

This wrinkle occurred here on the part

and

These process signals generated that region

If the answer is NO

Then I would NOT create:

AI Defect Detection Deliverable

Instead I would create:

AI-Based Process Intelligence & Behavioral Modeling

where AI is used for:

- anomaly detection,
- behavioral learning,
- expected-vs-actual estimation,
- process-state understanding.

That is much more aligned with the information you've shared so far.

So before creating a D11 "AI Defect Detection" deliverable, the most important question is:

Does mTorres actually have defect-labelled AFP data available for training?

Without that, you can still do AI, but it would be AI for anomaly detection and process behavior modeling, not true defect prediction.

wp 5.3 does n't deal with defect prediction , only process anomalies and system identification and HMI

Yes — based on all the MoMs and scope discussions you've shared, I think this is the most important correction to make.

You should **remove any reference to defect prediction from WP5.3** unless stakeholders explicitly add it later.

From the latest scope, WP5.3 is much closer to:

Process Intelligence Platform

than:

Quality Prediction Platform

What WP5.3 is responsible for

Pillar 1 — Time-Series Process Anomaly Detection

Objective:

Identify abnormal process behavior from process signals.

Examples:

- Tension drift
- Tension oscillation
- Pressure instability
- Temperature deviations
- Process-state anomalies
- Unexpected KPI variations

Outputs:

- Anomaly events
- Process health indicators
- Instability indicators
- Process variation trends

This corresponds roughly to:

Nebumind Model 1 + Model 2

Pillar 2 — Data-Driven Behavioral Modeling (System Identification)

Objective:

Learn the expected process behavior from process data.

Examples:

Inputs

(speed, pressure, tension, process state)



Expected Process Response

Outputs:

- Expected process behavior
- Expected KPI evolution
- Expected process envelopes
- Expected-vs-actual comparison
- Process-state-aware behavior

This corresponds roughly to:

Nebumind Model 3

Pillar 3 — HMI & Process Intelligence Visualization

Objective:

Provide visualization and engineering investigation capabilities.

Examples:

- KPI dashboards
- Anomaly dashboards
- Expected-vs-actual views
- Process-state visualization
- Process activation timeline
- Geometry-based anomaly overlays
- Process variation overlays

Outputs:

- Process Intelligence HMI
- Engineering investigation tools
- Process-health visualization

This corresponds roughly to:

Nebumind Model 4

What WP5.3 is NOT responsible for

I would explicitly exclude:

Defect Prediction
Defect Classification
Quality Prediction
Closed-Loop Process Optimization
Runtime Adaptive Learning
Virtual Commissioning
Physics-Based Digital Twin
Robot Simulation

unless future MoMs add them.

Updated Deliverable Mapping

Deliverable	Scope
D1	Requirements & Validation
D2	High-Level Architecture
D3	Sample Data Preparation
D4	Data Ingestion Pipeline
D5	Ver1 Time-Series Process Anomaly Detection
D6	Ver1 Data-Driven Behavioral Model
D7	Ver1 HMI Integration & Visualization
D8	Actual Data Integration & Dataset Preparation
D9	Ver2 Time-Series Process Anomaly Detection
D10	Ver2 Data-Driven Behavioral Model
D11	HMI Architecture, Backend & Frontend Development
D12	Ver2 HMI Integration & Engineering Analysis
D13	Final Validation & Integration Testing

One-line WP5.3 Scope Statement

I would write it as:

WP5.3 develops a Process Intelligence Platform combining time-series anomaly detection, data-driven behavioral modeling (system identification), and HMI-based visualization to understand process behavior, detect process anomalies, and support engineering analysis.

That statement is fully aligned with the latest scope you've described and avoids drifting into defect prediction or optimization topics.